Medical Delivery Quadcopter: Design Review

Battery check, system overview and comparative study

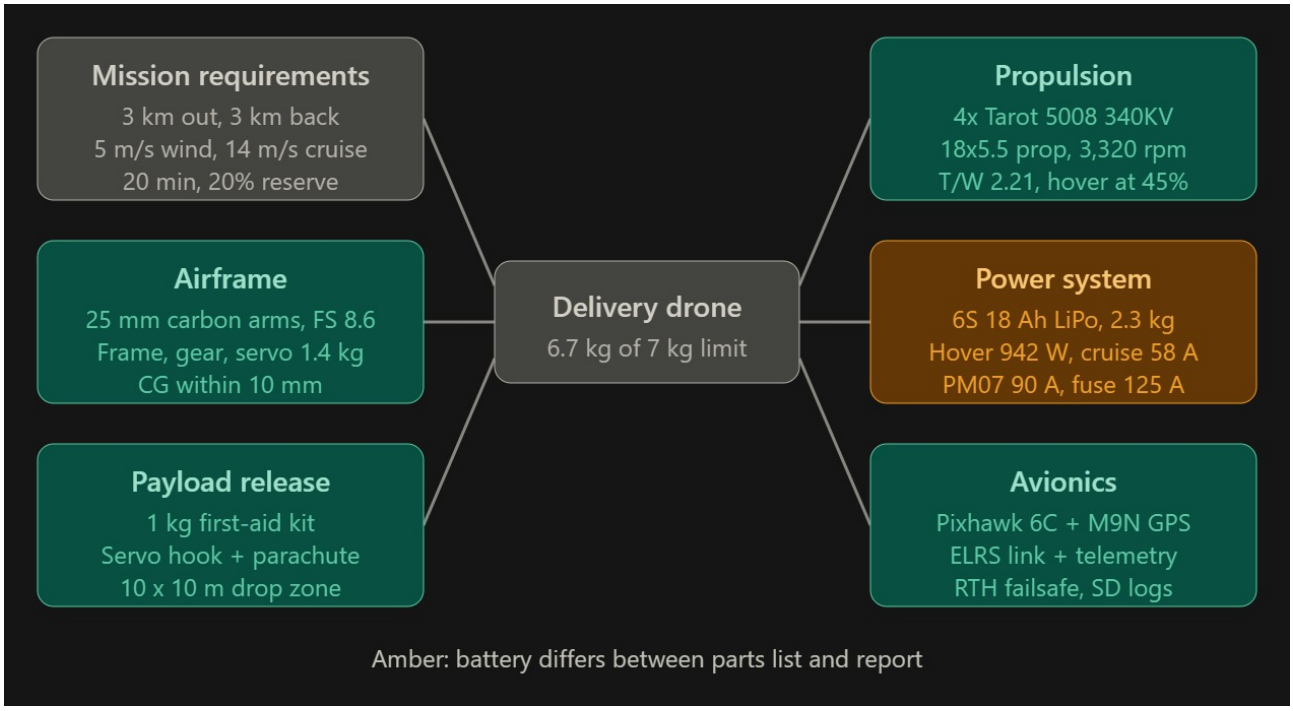
1. Battery Check: Report vs Parts List

Mission results for the report's 22.5 Ah pack compared with the 6S 18 Ah LiPo on the parts list.

Check	Report (22.5 Ah)	Parts list (18 Ah)
Usable energy	369 Wh	304 Wh
Hover endurance with margin	23.4 min	19.3 min (just under 20)
Energy left, worst-case wind	36%	22% (barely passes the 20% reserve)
Max one-way distance	3.81 km	3.08 km

2. System Mind Map: Components and Calculated Values

Each branch lists the selected components with the key calculated figures. Amber marks the battery that differs between the parts list and the report.



3. Comparative Study: Battery, Payload and Wind Sensitivity

Compares how battery capacity, payload, headwind and cruise speed affect hover endurance, usable energy left and maximum one-way distance (graphs use the report's 22.5 Ah, 21.6 V pack as baseline).

